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Bioenergy and biofuels: History, status, and perspective

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ABSTRACT

The recent energy independence and climate change policies encourage development and utilization of renewable energy such as bioenergy. Biofuels in solid, liquid, and gaseous forms have been intensively researched, produced, and used over the past 15 years. This paper reviews the worldwide history, current status, and predictable future trend of bioenergy and biofuels. Bioenergy has been utilized for cooking, heating, and lighting since the dawn of humans. The energy stored in annually produced biomass by terrestrial plants is 3–4 times greater than the current global energy demand. Solid biofuels include firewood, wood chips, wood pellets, and wood charcoal. The global consumption of firewood and charcoal has been remaining relatively constant, but the use of wood chips and wood pellets for electricity (biopower) generation and residential heating doubled in the past decade and will increase steadily into the future. Liquid biofuels cover bioethanol, biodiesel, pyrolysis bio-oil, and drop-in transportation fuels. Commercial production of bioethanol from lignocellulosic materials has just started, supplementing the annual supply of 22 billion gallons predominantly from food crops. Biodiesel from oil seeds reached the 5670 million gallons/yr production capacity, with further increases depending on new feedstock development. Bio-oil and drop-in biofuels are still in the development stage, facing cost-effective conversion and upgrading challenges. Gaseous biofuels extend to biogas and syngas. Production of biogas from organic wastes by anaerobic digestion has been rapidly increasing in Europe and China, with the potential to displace 25% of the current natural gas consumption. In comparison, production of syngas from gasification of woody biomass is not cost-competitive and therefore, narrowly practiced. Overall, the global development and utilization of bioenergy and biofuels will continue to increase, particularly in the biopower, lignocellulosic bioethanol, and biogas sectors. It is expected that by 2050 bioenergy will provide 30% of the world's demanded energy.

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Abbreviations: ABE, acetone–butanol–ethanol; FAME, fatty acid methyl esters

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1. Introduction

The annual terrestrial primary production adds approximately 120×10^{15} g of dry vegetative biomass [1], storing 2.2×10^{21} J of energy in plant materials [2]. The world energy demand was 5.5×10^{20} J in 2010. It is predicted to increase to 6.6×10^{20} J in 2020 and 8.6×10^{20} J in 2040 [3]. Largely, the bioenergy captured each year by land plants is 3–4 times greater than human energy demands. The bioenergy delivery potential of the world's total land area excluding cropland, infrastructure, wilderness and denser forests is estimated at 190×10^{18} J yr^{-1}, 35% of the current global energy demand [4].

Biofuels refer to plant biomass and the refined products to be combusted for energy (heat and light). Similar to fossil fuels, biofuels exist in solid, liquid, and gaseous forms. Human beings have been utilizing bioenergy and biofuels for domestic purposes since pre-recorded history. Wood was burned for heat and light to cook food, illumine night, warm shelters, and treat clay artifacts. Before the 19th century, wood was the predominant fuel for cooking and heating and plant oil was the chief fuel for lighting worldwide. Today, however, fossil fuels are the dominant energy sources, meeting $>80\%$ of the world's energy demand [5]. Nevertheless, fossil fuels are nonrenewable and their reserves are limited. At the current consumption rates, the supply of petroleum, natural gas, and coal will only be able to last for another 45, 60, and 120 years, respectively [6]. The dwindling supply and the soaring price of fossil fuels, especially petroleum, compel the world to develop renewable energy alternatives. Moreover, tremendous amounts of greenhouse gases have been released from fossil fuel consumption, elevating the atmospheric CO_2 concentration from the pre-industrial level of 280 ppm to the present nearly 400 ppm and causing disastrous climate change effects [7]. The incentives for mitigating global climate change further stimulate international communities to invest in development and utilization of renewable energy. Of the renewable energy sources, bioenergy draws major and particular development endeavors, primarily due to the extensive availability of biomass, already-existence of biomass production technologies and infrastructure, and biomass being the sole feedstock for liquid fuels. Currently, commercial production of electricity and transportation fuels from biomass feedstocks is practiced in most nations. The U.S. bioenergy production reached 4.76×10^{15} J (4.51 quadrillion Btu; 1 Btu = 1054.35 J) in 2011, accounting for 48.8% of the renewable energy and 5.8% of the total energy produced in the year [8]. Approximately 42% of the U.S. corn grains were used to generate 49 billion liters of bioethanol, representing 94% of the liquid biofuels produced in 2012 (52.2 billion liters) and replaced 10% of the nation's demanded gasoline fuel [9]. The U.S. Energy Independence and Security Act of 2007 mandates to increase annual biofuel addition to gasoline from 34 billion liters (9 billion gallons; 1 gal = 3.781 L) in 2008 to 136 billion liters by 2022, with 60 billion liters of the biofuel from lignocellulosic materials. A variety of conversion technologies to produce "drop-in" fuels like butanol and C_3–C_{10} hydrocarbons from wood and grasses are under investigation. The extensive production of biomass feedstock and biofuels, however, will generate significant environmental and socioeconomic impacts on soil, water, land, food, and civil development. This paper is to summarize the history of human utilization of bioenergy, review the present use and development of solid, liquid, and gaseous biofuels, and look into the future of bionenergy and biofuels. It aims to present a whole picture of bioenergy and biofuels and prepare readers to transit into the "after-fossil fuel" life.

2. Development and utilization of solid biofuels

2.1. Firewood

Wood and other plant materials have been directly burned for heating and cooking since the dawn of modern human beings. Before the discovery of fossil fuels, firewood was the primary fuel for domestic purposes. At 220–300 °C or higher temperature, most dry plant materials ignite in air to cause fire (flame), releasing the inherent bioenergy in heat and light. Combustion starts initially with exothermic pyrolysis of wood at around 260 °C to generate solid char and gaseous fume. Subsequently the char burns to ash and the fume to flame. Fire – the combustion of organic matter – is prompt oxidation of bio-carbon compounds by oxygen at high temperature and can be simply described as:

$$C_6H_{10}O_5 + 6O_2 \rightarrow 6CO_2 + 5H_2O + \text{heat} + \text{light} \quad (1)$$

It is not clear when human beings started the controlled use of fire for heat and light. The initial fire must be from lightening and volcanic eruption that ignited wood. Nevertheless, human beings had grasped the "bow-drill fire making" technique six thousand years ago [10]. Archaeologists identified charred animal bones and stone tools in wood ash in Wonderwerk Cave of Kuruman Hills in South Africa, providing evidence of controlled fire use by prehistoric "mankind" creatures one million years ago [11]. Wood, straw, hay, cattle dung, and peat have been intentionally collected, dried, and burned to prepare food and warmth. In the wilderness, bonfires and straw torches are commonly used for heating and lighting. Fire has also been used to clear land, to attack enemy in war, to smelt ores, and to treat clay artifacts for china, bricks, and tiles.

Firewood is normally packed in bundles and traded in volume. In the U.S., firewood is measured in "cord," with one cord equivalent to 128 ft^3 ($4' \times 8' \times 4'$) or 3.6 m^3. The weight of one cord firewood varies from 1350 to 2600 kg, depending on the wood type and moisture content. Hardwood is generally "heavier" than softwood. Green, unseasoned firewood may contain 150% (commonly 40–100%) moisture on the *dry mass basis* (The same applies in the following if not specified). Seasoned, air-dry firewood usually contains 10–25% moisture. "Wetter" firewood is normally lower in energy content, as certain heat has to be consumed to transform water into steam upon combustion. Therefore, green firewood should be seasoned for at least 6 months in a well-ventilated place to bring its moisture content to below 20% prior to use. The energy content of well-seasoned firewood is $\sim$15 MJ kg^{-1}, one-third to half of that of fossil fuels [12]. The energy would be fully released in heat and light upon complete combustion that produces only CO_2 and water vapor emissions. Unfortunately, combustion of wood and other raw plant materials in conventional furnaces is generally incomplete, resulting in significant smokes (a mixture of water vapor, volatile organic compounds, and carbon black particulates) and creosote (smoke condensate) that

are hazardous to human health, equipment, and the environment. Other minor emissions from burning firewood include carbon monoxide, methane, nitrogen oxides, and sulfur oxide [13].

Incomplete combustion of firewood is caused by insufficient oxygen, inadequate mixing, and/or low temperature [14]. The combustion efficiency of seasoned firewood in a typical burning device ranges from 80% to 85% [15]. Considering the 50–80% thermal transfer efficiency, conventional fireplaces, wood stoves, and boilers demonstrate an energy (thermal) efficiency of 40–68% [14]. The energy efficiency of a furnace is defined as:

$$\delta_{comb} = \frac{\text{thermal energy available in the flue gas}}{\text{chemical energy in the supplied fuel}} \quad (2)$$

For specially-designed wood stoves that include an additional hot-gas burner on the flame top, the combustion efficiency can reach 99% and the net energy efficiency 85%. The USEPA-certified wood stoves and fireplaces have an energy efficiency greater than 75% and fine particulate (PM2.5) emissions less than 0.6 g $(MJ)^{-1}$ (1.4 lb/million Btu) [16]. In developing countries, extension of energy-efficient firewood stoves (or firewood-saving stoves) has significantly reduced the consumption of firewood [17].

Firewood was the predominant fuel of the whole world for cooking and heating before the 19th century. In the U.S., wood-burning fireplace, heater, and cook stove were standard household equipment prior to the early 20th century. Even today, nearly 40% (2.6 billion) of the world population (mainly in the rural areas of developing countries in Asia and sub-Sahara Africa) relies on firewood to satisfy its energy requirement, with an annual consumption of 1730 million m^3 [5,18]. In 2012, India, China, United States, and Japan consumed 308, 185, 40, and 0.8 million m^3 of firewood, respectively [18]. Over the past thirteen years, the world consumption of firewood has slightly increased by 3%, while the share in the total energy consumption has been declining [18]. It can be projected that in the future the global consumption of firewood will remain relatively constant. For a specific nation or region, the firewood consumption will decrease with industrial advancement and stove renovation. Development of new technologies for improving combustion efficiency and energy efficiency of firewood furnaces will be the main trend.

2.2. Wood chips

Direct combustion of firewood instead of any refined or processed biofuels is the most efficient method to utilize bioenergy. Firewood, however, is bulky and not applicable to small, automated heating systems for controlled fuel value. In comparison, wood chips have the advantages.

Wood chips are small pieces ($1.3–7.6 \times 1.3–7.6 \times 0.3–0.6$ cm^3) of wood from cutting tree trunks and branches with a wood chipper. The material is commonly used for landscape mulching, playground surfacing, and wood pulp producing. Since the beginning of the 21st century, wood chips have been increasingly used for heating (bioheat) and electricity generation (biopower) [19]. Wood chips, however, are not used in rural areas of developing countries, as wood chippers are not available (affordable) to the residents. In the U.S., wood chips are actually less expensive and more available than firewood in that all parts of a tree can be efficiently processed into wood chips with an automated wood chipper and be conveniently transported, handled, and stored. To reduce the carbon footprint, many universities in the U.S. are shifting to automatically-fed wood chip boilers for heat, hot water, and electricity needs. For example, the woodchip boiler installed in Colgate University (Hamilton, NY) consumes 20,000 t of locally-produced wood chips per year, providing more than 75% of the heat and hot water that the campus demands [20]. The 0.5 MBtu h^{-1} wood chip boiler at Cayuga Nature Center, NY exhibits an average energy efficiency of 85% and reduces the heating cost by $15,000/yr over propane [21]. Co-firing has been commercially practiced by numerous coal-fired power plants to generate electricity, in which wood chips are combusted in 15–30 vol% mixtures with pulverized coal to drive steam turbines. The conversion efficiency of bioenergy to electricity in co-firing ranges 33–37% [19]. There are also 222 power plants directly fueled by wood chips in the U.S., producing annually 7.5 billion kW h of electricity [22]. In 2012, the U.S. generated 57.6 billion kW h of electricity from wood chips, accounting for 1.42% of the total produced electricity [3]. The world biopower generation capacity was 70 GW in 2010 and it is projected to increase to 145 GW by 2020 [23].

Compared with coal, wood chips emit much less sulfur oxides (SO_x) and nitrogen oxides (NO_x) upon combustion [24]. Nevertheless, wood chips may lose significantly in dry matter weight and energy value during storage [25]. Covered storage of pre-dried wood chips is economical for bioenergy recovery [26]. To minimize wood chip deterioration during storage, torrefaction of the material may be employed. Torrefaction is to heat wood at 240–300 °C in the absence of air for partial pyrolysis. The treatment can reduce wood chips by 20% in dry mass and consequently 10% in heating value, but results in a dry, hydrophobic, sterile, and stabilized product. Screening to remove finer pieces is also necessary prior to use of wood chips in automatically-fed boilers. As wood ash fuses to a greater extent than coal ash at temperature below 750 °C, boiler slagging and fouling can be a problem when wood chips are used in co-firing at rates higher than 30%. New technologies are currently under investigation to improve the performance and energy efficiency of biopower and bioheat using wood chips [27]. Given the present climate change policies, renewable portfolio standards, and environmental regulations, global production and utilization of biopower and bioheat from wood chips will increase steadily into the future [24].

2.3. Wood pellets

Relative to wood chips, wood pellets are a more processed biofuel product. Pellets are made by grinding wood chunks into sawdust through a hammer mill and subsequently compressing the sawdust through 6–8 mm holes in the die of a pelletizer. The high pressure raises the temperature of wood nuggets to plasticize the inherent lignin that glues the pellet as it cools. The pellets are 2–3 cm in length. The volumetric energy content of pellets can be increased from 9 to 18 MJ m^{-3} if the wood feedstock is completely torrefied before pelletization [28]. Grasses, crop residues, and nutshells can also be processed into pellets.

Wood pellets have a low moisture content in the range of 5–10% and a high packing density at around 650 kg m^{-3}. High quality pellets are mechanically durable ($< 2.5\%$ broken into finer particles after each handling), have a light brown color, and possess less than 0.7% mineral ash and $< 1\%$ fine powders [29]. The smooth, cylindrical shape and the small size of wood pellets allow the biofuel to be fed automatically at fine calibration. The grain-like geometry, high density, and low moisture content of the material further permit compact storage, bunker transfer, and long-distance transport. Wood pellets, however, are more costly than wood chips and therefore, their use is limited primarily in residential heating of developed countries. In countries like China, Japan, Germany, U.K., and Netherlands, wood pellets are also used for biopower generation. The current U.S. market price of wood pellets is roughly $5 per bag (18 kg or 40 lb) or $250 per ton. Automatically-fed pellet stoves are used to burn wood pellets for space heating. The energy efficiency of such a stove ranges from 70% to 83% [30]. The appliance has a fuel hopper to contain 16–60 kg pellets that are adequate for one day operation. An auger feeder conveys a few

pellets at a time into the combustion chamber for burning. More advanced models are even equipped with a small computer and thermostat to govern the pellet feed rate.

The global production of wood pellets was 19.1 million tons in 2012. The production is projected to increase to 30.2 million tons in 2015 and further to 45.2 million tons in 2020 [31]. The present U.S. wood pelletization capacity is estimated at 6 million tons per year. In 2012, the U.S produced 4.1 million tons of wood pellets and increased the consumption to 2.8 million tons (from 1.8 million tons in 2008) [18]. The European countries consumed 13 million tons of wood pellets in 2012, of which 1.3 million tons were imported from the U.S., making U.S. the largest wood pellets exporter of the year [32]. The rapid expansion of the U.S. wood pellet production has brought severe environmental concerns to the nation, as more forests and wetlands are being cleared.

2.4. Charcoal

Compared with fossil fuels, wood fuels (firewood, wood chips, and wood pellets) are generally lower in energy content but higher in combustion emissions. Furthermore, the typical temperature of wood fire is below 850 °C, unable to melt many metals [33]. To overcome these drawbacks, human ancestors developed pyrolysis techniques to transform wood into charcoal, a carbon-enriched, porous, greyish black solid. Heating wood materials in a kiln or retort at around 400 °C in the absence air until no visible volatiles are emitted yields high quality charcoal. The pyrolysis reaction can be illustrated by the equation [34]:

$$C_6H_{10}O_5 \rightarrow 3.75CH_{0.60}O_{0.13}\ \text{(charcoal)} + 2.88H_2O + 0.5CO_2 + 0.25CO + C_{1.5}H_{1.25}O_{0.38}\text{(tar)} \quad (3)$$

The yield of charcoal is approximately 35% of the original wood dry mass. Good quality charcoal has an energy content of 28–33 MJ kg^{-1}, higher than that of coal. It burns without flame and smoke, giving a temperature as high as 2700 °C [34]. As such, wood charcoal is commonly made into briquettes for barbecuing. To prepare barbecue charcoal briquettes, small amounts of anthracite coal, mineral charcoal, starch, sodium nitrate, limestone, and borax, and sawdust are generally added to sawdust-derived charcoal. The additives act as binders, improve ignition, promote steady burning, and make manufacturing more efficient [35].

In the “Bronze Age” back to 3000 B.C., human beings started to use charcoal in metallurgy to smelt ores for copper and iron. Charcoal was the designated governmental fuel for cooking and heating in China’s Tang Dynasty in 700 A.D. During 1931–1960 in China and the World War II in Europe when gasoline was scarce, many automobiles were powered by wood gas, a mixture of carbon monoxide and hydrogen generated by partially burning charcoal in a gasifier [36]. Today charcoal is still a valuable product used for cooking, barbecuing, heating, air and water purification, art drawing, and steel-making.

The global wood charcoal production amounted 51 million tons in 2012, increasing by 5% from the 2008 figure. Around 31 million tons were produced in African countries. In 2012, Brazil, India, China, U.S., and Russia produced 7.6, 2.9, 1.7, 0.85, and 0.053 million tons of charcoal, respectively [18]. Predicting from the past-five year trend, the world annual production and utilization of wood charcoal will remain relatively constant at around 50 million tons.

3. Development and utilization of liquid biofuels

3.1. Bioethanol

Gasoline, a volatile liquid of C_4–C_{12} hydrocarbon mixture produced by cracking crude oil (petroleum), is the predominant fuel to power petrol engine automobiles. One barrel (42 gal) crude oil can be refined to produce 19 gal of gasoline [37]. Today there are over 600 million passenger cars traveling the streets and roads of the world, daily consuming 930 million gallons of gasoline. The U.S. daily consumption of gasoline was 366 million gallons in 2012 [37]. The world’s passenger transportation relies on crude oil for adequate supply of gasoline. The heavy dependence on crude oil, however, has generated tremendous environmental and socioeconomic impacts. One example is the 1970s Energy Crisis, during which the economies of major industrial countries including United States, Canada, Western Europe, Japan, and Australia were severely affected by substantial shortage and soaring price of crude oil due to OPEC oil export embargo (in 1973) and Iranian Revolution (in 1979) as well as the growing citizens’ environmental concerns [38]. Furthermore, the supply of crude oil will only be able to last for another 45 years at the current consumption rate [6]. It is urgent to develop renewable sources of gasoline substitute that fits the existing liquid fuel supply systems.

Biomass is currently the only renewable feedstock material to produce liquid fuel. One commercially-practiced technology is to produce ethanol (alcohol) by fermenting plant biomass-derived simple sugars (i.e., glucose, fructose, and other monosaccharides). Ethanol can be used as a gasoline substitute to power petrol engines. The fermentation technology dated back to 4000 B.C., by which human beings make alcohol as a drink from berries, grapes, honey, and cereals. As a matter of fact, ethanol was tested as an engine fuel far before the commercial production of gasoline in 1913. Early in 1826, the American inventor Samuel Morey designed an internal combustion engine that was fueled by ethanol and turpentine to run a boat at 7 to 8 mph (miles per hour). In 1860, the German engineer Nicolaus August Otto developed another internal combustion engine that ran on an ethanol fuel blend [39]. Later the American industrialist Henry Ford constructed tractors that could be powered by ethanol. The obstacle that prevented ethanol from being used as an engine fuel in the U.S. was the alcohol tax enacted in the 1860s to fund the Civil War [40]. Even when gasoline began to prevail in the late 1910s, many scientists advocated the competence and sustainability of ethanol in the fuel industry. In 1917, Alexander Graham Bell highlighted the abundance of potential feedstocks for the production of ethanol: “any vegetable matter capable of fermentation, crop residues, grasses, farm waste, and city garbage.” An article in a 1918 issue of *Scientific American* commended the effectiveness of a fuel blend of 25% gasoline, 25% benzole, and 50% alcohol, proposing it as a solution to the oil reserve-diminishing problem [39]. A revival of interest in ethanol as a fuel was brought forth by the 1973 oil crisis. Substantial research efforts were made to develop biotechnical conversion of plant biomass to ethanol. These attempts, however, did not finalize the development of cost-effective, industry-applicable cellulose enzymes that would effectively hydrolyze plant cellulose to simple sugars for ethanol fermentation [41]. The first pilot bioethanol plant with a distillation column was established at South Dakota University (Brookings, SD) in 1979 [39].

Bioethanol is ethanol produced from vegetative biomass through fermentation, in which the following biochemical reactions are involved:

$$(C_6H_{10}O_5)_n\ \text{(starch, cellulose, sugar)} + nH_2O \rightarrow nC_6H_{12}O_6\ \text{(glucose, fructose)} \quad (4)$$

$$(C_5H_8O_4)_n\ \text{(hemicellulose)} + nH_2O \rightarrow nC_5H_{10}O_5\ \text{(xylose, mannose, arabinose, etc.)} \quad (5)$$

$$C_6H_{12}O_6 \rightarrow 2CH_3CH_2OH\ \text{(ethanol)} + 2CO_2 \quad (6)$$

$$C_5H_{10}O_5 \rightarrow 5CH_3CH_2OH\ \text{(ethanol)} + 5CO_2 \quad (7)$$

Theoretically all plant materials can be used to generate bioethanol, as cellulose, hemicellulose, and lignin are the three major construction compounds interwoven with one another to form cell walls of all plants (Fig. 1). Nevertheless, it is fairly challenging to "unbraid" these three compound fibrils and depolymerize cellulose and hemicellulose into simple sugars (Eqs. (4) and (5)). Research has been intensively conducted to develop effective "pretreatment" methods for obtaining simple sugars from lignocellulosic materials.

Commercial bioethanol is currently produced from starch/sugar-based crops including sugar cane, sugar beet, sweet sorghum, corn, wheat, barley, potato, yam, and cassava. The global commercial bioethanol production was 4.0 billion gallons in 1990. It increased slightly to 4.5 billion gallons in 2000 but rapidly to 23.3 billion gallons in 2010. In 2013 the figure was 23.4 billion gallons, with 56.8%, 26.7%, 5.9%, 3.0%, and 2.1% of the production in the U.S., Brazil, Europe, China, and Canada, respectively [42]. During 2010–2013, the annual U.S. bioethanol consumption amounted roughly 13 billion gallons. In fact, the U.S. invested 42% of its harvested corn grains (114 million tons/yr) in bioethanol production [43], attempting to replace 10% of its gasoline demand with bioethanol. Sugar cane is the predominant feedstock for bioethanol in Brazil. The European countries use primarily wheat and sugar beet to produce bioethanol. In China, the principal bioethanol feedstocks are corn, wheat, and cassava, while in Canada, they are corn and wheat.

The typical procedure for manufacturing bioethanol from food crops is outlined in Fig. 2. It involves a number of steps such as milling, liquefication, saccharification, fermentation, distillation, drying, and denaturing. Given corn as the feedstock, corn kernels are ground to 3–4 mm flour, wetted with water into slurry, and "cooked" (heated by steam at 110–120 °C for 2 h) into corn mash. The enzymes α-amylase and glucoamylase are added during the following liquefication (80–90 °C) and saccharification (30 °C) stages, respectively. The saccharified corn mash is then fermented at 32 °C for 50 h with addition of yeast, yielding 8–12% ethanol by weight in the slurry. Ethanol is recovered from the slurry by distillation, de-watered by passing through molecular sieves, and denatured by mixing with 2–5% gasoline [44]. Typically, 2.5–2.9 gal of bioethanol can be generated from 1 bushel (25.4 kg) corn grains.

Bioethanol made from food crops is viewed as "first-generation" biofuel, which competes with animal feed and human food for the source materials. To minimize the adverse impacts, manufacturing "second-generation" bioethanol from non-food lignocellulosic plant materials has been explored. Indeed, lignocellulosic materials are widely available: forest slashes, crop residues, yard trimmings, food processing waste, and municipal organic refuses can be the feedstock for bioethanol. Dedicated biomass crops such as switchgrass, miscanthus, giant reed, energy cane, napier grass, grain sorghum, shrub willow, and hybrid poplar can also be grown on marginal land for the purpose [45]. Lignocellulosic bioethanol production involves three categories of costs: the costs of feedstock, the costs of sugar preparation, and the costs of ethanol production. Among these three categories, conversion of cellulosic components into fermentable sugars is the major technological and economical bottleneck. Research has been focusing on development of cost-effective techniques for extracting simple sugars from lignocellulosic biomass. Two feedstock treatment technologies have been proposed: acid hydrolysis and enzymatic hydrolysis [46]. Acid hydrolysis is to treat pulverized plant mass with an acid solution (e.g., sulfuric acid) to help sugar release. If a dilute acid (e.g., 1–10% H_2SO_4) is used, a high temperature (e.g., 237 °C) and high pressure (e.g., 13 atm) environment is necessary. The treatment is rapid (e.g., less than 15 s) and can be in a continuous flow, but the sugar yield is around 50% (of the cellulose-based figure). The low sugar recovery is due to rapid degradation of the produced sugars to furfural and other chemicals under high temperature and pressure conditions (Fig. 1). Considering that 5-carbon sugars (pentoses) are degraded more rapidly than 6-carbon sugars (hexoses), a two-stage acid hydrolysis process can be employed to decrease sugar degradation: plant biomass is initially subject to mild conditions (e.g., 135 °C) to recover pentoses and then experiences harsher conditions to release hexoses [47]. The ethanol yield per dry ton wood by this

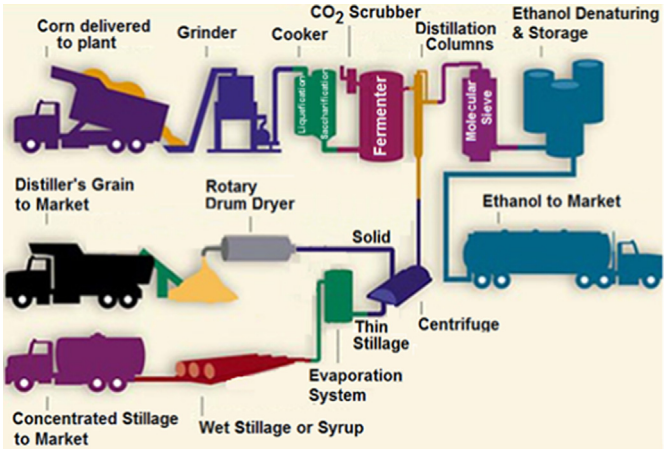


Fig. 2. The technical flow chart of bioethanol production from corn in the U.S. (Source: Renewable Fuels Association)

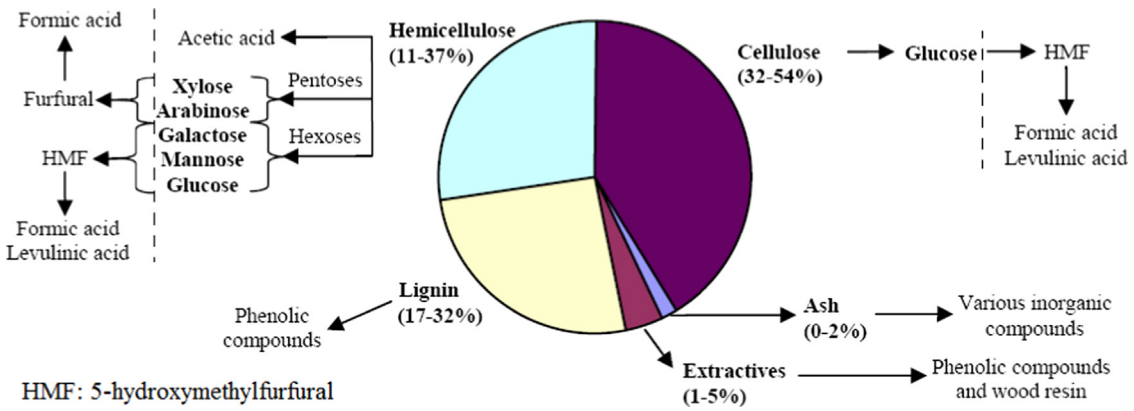


Fig. 1. Composition of lignocellulosic materials and their potential hydrolysis products and further degradation compounds (chemicals behind the dashed line) [102].

method is approximately 55 gal [46]. If a concentrated acid (e.g., 30–40% H_2SO_4) is used, lignocellulosic materials are initially treated with a dilute acid under mild conditions (e.g., 100 °C) for 2–6 h to recover pentoses. The solid residue is then rinsed by water, press-drained, and soaked in the concentrated acid solution at 100–150 °C for 2–4 h. The reactor contents are then filtered to remove lignin and may be recycled to the dilute acid hydrolysis reactor to reuse the acid. The overall sugar recovery by this method can reach by 80%. The process is slow, however, and the acid is difficult to separate from the sugar solution. Without acid recovery, lime must be used to neutralize the sugar solution prior to fermentation, creating additional costs and calcium sulfate waste. By this method 60–70 gal of ethanol can be produced per dry ton of corn stover [46]. In enzymatic hydrolysis, active enzymes are added to decompose lignocellulosic materials into simple sugars. As cellulose in plant materials exists in well-organized crystalline structure and is protected by hemicellulose and lignin layers, it is highly resistant to enzymatic attack. Pretreatment, such as freezing, radiation, steam explosion, or autohydrolytic hydrothermal deconstruction, is necessary to initially degrade hemicellulose and solubilize lignin [48]. Glycoside hydrolase and carbohydrate esterase enzymes can readily liberate fermentable sugars from pre-treated lignocellulosic substrate. Research has been intensively conducted to complement the commercial preparation of these cellulases. It was discovered that the fungus *Trichoderma reesei* could produce cellobiohydrolases (exo-β-1,4-glucanases) and Cx-cellulases (endo-β-1,4-glucanases) that are effective in depolymerizing cellulose. This fungus contains 200 glycoside hydrolase genes that control the production of at least 10 different enzymes necessary for complete hydrolysis of lignocelluloses. The protein structure of these enzymes was well studied. Substantial efforts have been carried out to improve the major proteins in the enzyme preparations [41]. Thermostable cellulases that retain high cellulose-degrading activity at >70 °C were also identified from several mesophilic and thermophilic fungal *strains Talaromyces emersonii*, *Thermoascus aurantiacus*, *Chaetomium thermophilum*, *Myceliophthora thermophila* and *Tribulus terrestris*. Application of these enzymes in cellulosic bioethanol production is under experiment. In general, the costs for preparing celluloytic enzymes have been significantly reduced during the past decades, yet it is still the major obstacle for commercial-scale enzymatic hydrolysis processes. In the meanwhile, new yeast strains for more effective ethanol fermentation have been developed [49]. Using advanced technologies in biomass pretreatment and acid/enzymatic hydrolysis, pilot cellulosic ethanol plants operated by a number of companies (e.g., Abengoa Bioenergy, Salamanca, Spain; Iogen Energy, Ottawa, Canada; and Poet, LLC., Sioux Falls, SD, USA) have achieved the ethanol productivity of 68–83 gal t^{-1} biomass.

Thermochemical transformation was also studied for producing bioethanol from lignocellulosic materials. Plant biomass is first gasified to produce syngas (a mixture of H_2 and CO). The syngas is then introduced into a specially-designed fermenter where particular microorganisms utilize the syngas for energy and generate ethanol. Alternatively, the syngas is passed through a reactor and is transformed to ethanol in the presence of chemical catalysts ($3CO+3H_2 \rightarrow CH_3CH_2OH+CO_2$). Ethanol yields up to 50% of the syngas mass have been obtained using the thermochemical transformation methods [46]. Commercial scale plants to produce ethanol from syngas have not entered into practice, likely due to the high technological complexity and low economic viability.

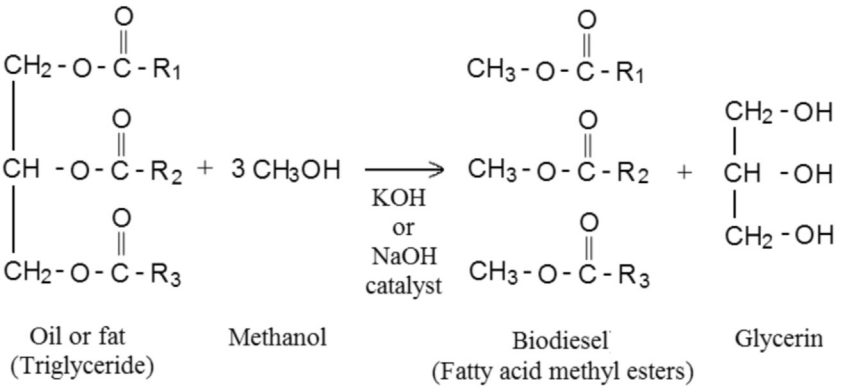
Gasoline blended with 10 volume% bioethanol (E10) is now ubiquitous on the U.S. fuel market [50]. The bioethanol, however, is predominantly from corn. Commercial production of bioethanol from lignocellulosic materials has not started in the nation, largely due to the low profitability with the current conversion technology and feedstock supply system. The Energy Independence and Security Act of 2007 regulates that the U.S. reduces its gasoline consumption by 20% within 10 years and increases biofuel addition to gasoline from 4.7 billion gallons in 2007 to 36 billion gallons in 2022, with 21 billion gallons of biofuels (bioethanol 18.15 billion gallons) from non-corn products. Accordingly, commercial production of bioethanol from lignocellulosic feedstocks should start in 2010, with targets of 0.5 billion gallons in 2012 and 1.7 billion gallons in 2013. In fact, the nation produced merely 20,000 gal of cellulosic ethanol in 2012 and 218,000 gal in 2013 through pilot plant operations [51]. Worldwide, the first commercial-scale cellulosic ethanol plant (The Crescentino Bio-refinery, Crescentino, Vercelli, Italy) entered into full operation on October 9, 2013. The plant is running by the Italian company Beta Renewables to annually produce 20 million gallons of ethanol from wheat straw and giant reed using the patented "Proesa" technology in which plant biomass is pre-treated with steam (high temperature and pressure), followed by enzymatic hydrolysis. Beta Renewables is going to build another 80-million-gallon cellulosic ethanol plant in Fuyang, Anhui, China. The first U.S. commercial cellulosic ethanol plant (Indian River BioEnergy Center, Vero Beach, FL. Owned by the Swiss firm Ineos Bio) with the designed 8 million gallon annual capacity is at the testing stage to produce bioethanol from vegetative residues using gasification and anaerobic fermentation. Official operation is expected in 2014 [52]. A few more commercial scale cellulosic ethanol plants are under construction in the U.S., including Abengoa Cellulosic Ethanol Biorefinery (Hugoton, KS. 25 Mgal/yr), Bluefire Biorefinery (Fulton, MS. 18 Mgal/yr), and Poet Liberty (Emmetsburg, IA. 25 Mgal/yr). It is clear that commercial production of cellulosic ethanol will speedily expand in the near future, boosting the global production and utilization of bioethanol as a gasoline alternative.

3.2. Biodiesel

Diesel is a C_8–C_{25} hydrocarbon liquid fuel produced from petroleum by fractional distillation at 200–350 °C. One barrel (42 gal) of crude oil can generate 10 gal of diesel [37]. The fuel has an energy content of 38.3 MJ L^{-1}, higher than that of gasoline (34.7 MJ L^{-1}) [53]. Diesel is an essential fuel for diesel engines and an important fuel for home heating in developed countries. Most heavy-duty vehicles and machines are powered by diesel engines, such as tractors, trucks, construction machineries, mining equipment, and military carriers. Due to different ignition mechanisms, diesel engines cannot use gasoline as the fuel, and versus versa. The average daily consumption of distillate fuel oil (diesel and fuel oils) in the U.S. was 107 million gallons in 2012. In the same year, the world daily production of distillate fuel oil amounted 1057 million gallons [3]. Given the world's dwindling petroleum reserves, renewable fuel sources alternative to petrol diesel are warranted.

Biodiesel, a yellowish liquid derived from vegetable oil, animal fats, algal lipids, or waste grease through "transesterification" in the presence of alcohol and alkaline catalyst (Eq. (8)), has been commercially produced and used a petrol diesel substitute. In chemical structure, biodiesel is mono-alkyl esters of fatty acids. Though specific properties vary with the feedstock lipids, biodiesel demonstrates a specific gravity 0.873–0.884, kinematic viscosity 3.8–4.8 mm s^{-2}, cetane number 50–62, cloud point −4–14 °C, and flash point 110–190 °C. Its energy

density (high heating value) is 38–45 MJ kg^{-1}, approximately 90% of that of petrol diesel [54].



(8)

Here R_1, R_2, and R_3 are different or the same aliphatic hydrocarbon groups.

Vegetable oil and animal fats were widely used in oil lamps for lighting before the invention of gas lights and electric lights in late 18th century, but for making biodiesel it did not occur until 1930s [55]. When the German engineer Rudolf Diesel invented the compression-ignited diesel engine in 1893, he envisioned that pure vegetable oil could power the machine for agriculture in remote areas of the world. He carried out extensive tests on various vegetable oil fuels, believing that farmers could benefit from producing their own fuel. At the 1900 World's Fair in Paris, the French Otto Company demonstrated a diesel engine that was fueled by peanut oil. The attention for developing vegetable oil-based fuels, however, was quenched by the later widespread availability and low price of petrol diesel. Despite this, interest in vegetable oil to power internal combustion engines still existed in a number of nations. Experiments in Belgium, Germany, Italy, France, Japan, China, Argentina, and other countries recognized the high viscosity of vegetable oils as the key problem when used to fuel diesel engines. Various methods were tested to overcome this drawback, such as pre-heating the vegetable oil, blending it with petrol diesel or ethanol, and thermochemically cracking the oil. In 1937, the Belgian scientist George Chavanne patented the "Procedure for the transformation of vegetable oils for their uses as fuels," presenting the transesterification (alcholysis) method for breaking down triglyceride molecules (oil and fat) by replacing the glycerin moiety with ethanol or methanol (Eq. (8)). This is the earliest description of biodiesel generation [56]. In 1977, the Brazilian scientist Expedito Parente developed the first industrial process for the production of biodiesel. In 1989, the world's first industrial-scale biodiesel plant was operated in Asperhofen, Austria to produce biodiesel from rapeseed. The U.S. started its commercial operation (Pacific Biodiesel, Maui, Hawaii) in 1996 to process waste grease into biodiesel. The historically high petroleum prices after 2001 and the increased awareness of energy security promoted biodiesel to a popular fuel on the global fuel market. In 2013, the world biodiesel production reached 6289 million gallons [57].

Vegetable oils, algal lipids, animal fats (beef tallow, pork lard, chicken fat, etc.), and yellow grease (used vegetable oil) can all be used as feedstock to produce biodiesel. Common oil seed plants include camelina, canola, castor bean, coconut, jatropha, palm, peanut, rapeseed, soybean, sunflower, and tung. The oil productivity of soybean is approximately 194 gal ha^{-1} given the typically 3363 kg ha^{-1} yield of seeds with 20% oil content. For canola, the oil productivity is around 350 gal ha^{-1}. In recent years, selected algae species were also studied for cultivating as a biodiesel feedstock. The strains with high oil contents (e.g., $>20\%$) and satisfactory biomass potential (e.g., >20 dry t ha^{-1} yr^{-1}) include *Chaetoceros calcitrans*, *Skeletonema costatum*, *Phaeodactylum tricornutum*, *Chlamydomonas reinhardtii*, *Calluna vulgaris*, *Dunaliella salina*, *Dunaliella teriolecta*, *Scenedesmus obliquus*, and *Neochloris oleabundans*. Critical challenges still exist in cost-effective cultivation of algae, in energy-efficient harvest of algal biomass, and in effectual algal oil extraction.

The common technical procedure of biodiesel production is presented in Fig. 3. Moisture-free vegetable oil is pre-heated to 50–60 °C. A "methoxide" mixture of methanol (20% volume of the oil) and sodium hydroxide (5 g L^{-1} oil) is added to the oil in a closed reactor to start transesterification (Eq. (8)). If yellow grease is the feedstock, the material has to be pre-treated to remove any water and particulates, and undergo "acid-catalyzed esterification" with methanol to eliminate free fatty acids prior to the base-catalyzed transesterification (Fig. 3). After 2 h of transesterification at 50–60 °C under mechanical stirring, the mixture is allowed to settle at room temperature for 2–12 h. The bottom crude glycerin layer is then separated from the top crude biodiesel layer by funnel drain. The crude biodiesel contains small amounts of methanol, soap, and mono/di/triglycerides. These impurities need to be removed prior to fuel use of biodiesel. Purification is generally achieved by water washing and drying or by membrane refining [58]. The biodiesel yield from the transesterification reaction is typically the same volume as the feedstock oil. The reaction conditions and purification operations, however, greatly

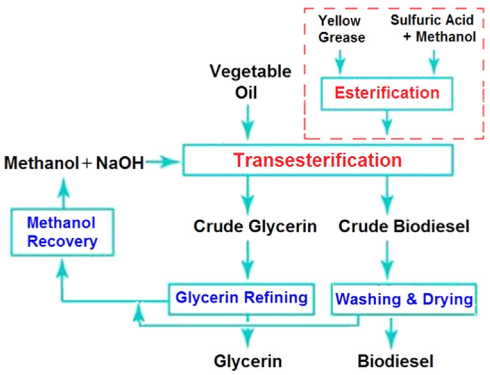


Fig. 3. The technical flow chart of biodiesel production from (used) vegetable oil

influence the yield and quality of the final biodiesel products. In the U.S., the quality of biodiesel is regulated by the ASTM D6751 (Standard Specification for Biodiesel Fuel Blend Stock (B100) for Middle Distillate Fuels). In European countries, it is by the EN-14214 (Automotive fuels – Fatty acid methyl esters (FAME) for diesel engines – Requirements and test methods).

Over the past decade, signification research progress has been achieved to optimize the biodiesel production procedure (transesterification), to determine the feedstock oil characteristics that control the biodiesel quality, to identify plant and algal species that have high oil yield potential, and to develop value-added utilization of the crude glycerin. By reviewing the published literature, Hoekman et al. [54] concluded general guidelines of feedstock selection for producing biodiesel that meets the two critical fuel quality desires "(1) low temperature performance and (2) oxidative stability. For good low temperature performance, biodiesel should have low concentrations of long-chain saturated FAME. For good oxidative stability, biodiesel should have high concentrations of saturated and monounsaturated FAME, but low concentrations of multi-unsaturated FAME."

The world production of biodiesel has increased steadily from 213 million gallons in 2000 to 6289 million gallons in 2013 [57]. The top five biodiesel production countries are the European Union (mainly Germany, France, Spain, Italy, and Poland), Brazil, Argentina, the U.S., and China, using oils from soybean, rapeseed, canola seed, sunflower seed, castor bean, animal fats, and yellow grease. The U.S. produced 1339 million gallons of biodiesel in 2013, solely from soybean oil [58]. Currently B20 (20% biodiesel, 80% petroleum diesel) is the most common biodiesel blend in the U.S. With the world's growing demand for biofuels, global production and consumption of biodiesel will maintain consistent increase. The advantages of biodiesel also include a cleaner emission profile, production simplicity, ease of use, and cost-competitiveness. The promising future of biodiesel, however, also lies in the world's ability to economically produce renewable feedstocks such as vegetable oils and algal lipids in sustainable manners, without supplanting land for food production.

3.3. *Pyrolysis bio-oil*

Fossil fuels were formed from remains of plants and animals millions of years ago under the work of geothermal heat and pressure. Attempting to produce similar fuels, human beings treated vegetative biomass in a simulated high temperature, high pressure, and oxygen-free environment. Yet the products were little like fossil fuels.

Pyrolysis is to heat plant biomass at 300–900 °C in the absence of air. The technique was used in ancient China to prepare charcoal and by indigenous Amazonians to generate biochar three to five thousand years ago. Pyrolysis of plant materials results in three products: biochar (the black, solid residue), bio-oil (the brown vapor condensate), and syngas (the uncondensable vapor) (Fig. 4). There are generally two types of pyrolysis techniques: slow pyrolysis and fast pyrolysis, referring to the heating rate [59]. Slow pyrolysis is to heat organic residues in batch reactors at 300–600 °C in the absence of air for a number of hours or days. The typical yields of biochar, bio-oil, and syngas in slow pyrolysis are 35%, 30%, and 35% of the dry feedstock biomass, respectively [60].

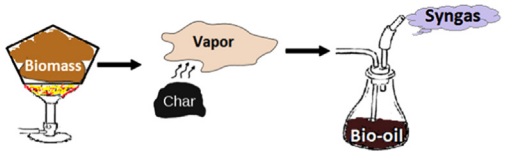


Fig. 4. Pyrolysis of plant biomass to generate char, bio-oil, and syngas.

Fast pyrolysis is to rapidly raise the temperature of dried and ground plant biomass (< 2 mm) to approximately 500 °C within 2 s in a continuous flow system in the absence of oxygen. Typical yields of biochar, bio-oil, and syngas in fast pyrolysis are 10-30%, 50–70%, and 15–20% of feed biomass, respectively [61]. Most organic residues can be used to produce pyrolysis bio-oil, such as wood, switchgrass, crop straw, sawdust, sugar cane bagasse, peanut hulls, and poultry litter. However, materials low in nitrogen and ash contents are preferred; wood is the common feedstock for quality bio-oil [59]. By manipulating the temperature, duration, and oxygen availability of pyrolysis, it is possible to optimize for one or more of the three products. Up to 75% of the feedstock biomass (and its energy) can be converted to bio-oil by optimized fast pyrolysis [62].

Crude pyrolysis bio-oil contains significant contents of colloidal char particles and water and differs significantly from petrol distillate fuels in physical, chemical, and energy properties (Table 1). More than 300 compounds were identified in pyrolysis bio-oil as acids, alcohols, aldehydes, esters, ketones, sugars, phenols, sugars, furans, guaiacols, syringols, alkenes, aromatics and nitrogen compounds [59,63]. Due to its high moisture content and acid content, crude pyrolysis bio-oil is instable, corrosive, viscous, low in energy density, immiscible with hydrocarbon fuels, and difficult to ignite [64,65]. Crude bio-oil can be directly burned using standard atomization techniques in industrial scale combustion systems, provided that the burner is set up to accommodate the unique characteristics of the liquid [59]. Nevertheless, significant upgrading is needed before the liquid can be utilized as a petrol distillate fuel alternative. The upgrading is to reduce the moisture content and acidity of bio-oil and improve its heating value and storage stability. Over the past decades, a number of upgrading techniques have been investigated, including hydrogenation, hydrodeoxygenation, esterification, catalytic cracking, molecular distillation, supercritical fluidization, emulsification, steam reforming, and blending [66,67]. Moderately upgraded bio-oil has been applied to substitute for heavy fuel oils (e.g., diesel and No. 2 heating oil) to power static appliances including boilers, furnaces, engines, and electric generators. Economically feasible, industry-applicable techniques for upgrading bio-oil into transportation fuels, however, are still under development. Pyrolysis bio-oil also serves as a potential feedstock for valuable chemicals, preservatives, lubricants, binders, paints, thickeners, stabilizers, etc. Yet with the present technologies, recovery of pure compounds from bio-oil is not economically unattractive [59].

Globally, a number of pilot plants have been operating at up to 200 kg h^{-1} feedstock processing capacity to convert woody biomass into bio-oil. These plants include DynaMotive, PyroVac Inc., Canada; ENEL, Italy; Ensyn, USA; Bioware Technologia, Brazil; BTG,

Table 1
Properties of wood-derived pyrolysis bio-oil as compared with petroleum distillate fuels [64].

Properties	Pyrolysis bio-oil	Petroleum distillate fuel
Water content (wt%)	15–30	0.1
pH	2–3	–
Density (kg m^{-3})	1.2	0.94
Elemental composition (wt%)		
C	54–58	85
H	5.5–7.0	11
O	35–40	1.0
N	0–0.2	0.3
Ash	0–0.2	0.1
High heating value (MJ kg^{-1})	16–19	40
Viscosity (50 °C) (cSt)	33–83	190
Solid particulates (wt%)	0.2–1	1
Distillation residue (wt%)	Up to 50	1

Ecosun, The Netherlands; PYTEC, PKA, Chemviron Carbon, Germany; DynaMotive, Compact Power, UK; Lambiotte, Belgium; Novasen, Senegal; and BEST Energies, Australia [59]. The resulting bio-oil is used as heating fuels and as industrial feedstock materials. Driven by the increasing interest in renewable fuels, pyrolysis bio-oil has a promising future. Nevertheless, commercial production and utilization of pyrolysis bio-oil as a petrol fuel alternative is still facing major technological challenges [68].

3.4. Drop-in biofuels

On the volume basis, bioethanol has an energy density 67% of that of gasoline, while biodiesel 90% of that of petrol diesel. More important, bioethanol and biodiesel demonstrate higher oxygen content and greater dissolution capability than petrol fuels and therefore, are more corrosive to engines and fuel storage and distribution equipment [69]. At blending rates above 20%, biofuel blends can damage elastomer and metallic parts of engines, storage tanks, and dispensers, threatening the infrastructure compatibility of the currently existing fuel supply system.

Drop-in biofuels are biomass-derived liquid hydrocarbons that meet the existing petrol distillate fuel specifications and be ready to "drop in" to the existing fuel supply and use infrastructure. Presently at the research and development stage, drop-in biofuels have the features to minimize the infrastructure and engine compatibility issue. Drop-in biofuel candidates include butanol, liquefied biomass, sugar hydrocarbons, syngas complexes, and others [70]. These biofuels follow different production pathways, such as fermentation of lignocellulosic sugars, catalysis of lignocellulosic sugars, hydropyrolysis of biomass, hydrothermal biomass liquefaction, bioethanol transformation, and syngas upgrading. Butanol can be produced by the acetone–butanol–ethanol (ABE) fermentation process using the solventogenic clostridial bacteria strains (e.g., *Clostridium acetobutylicum* EA2018 and *Clostridium beijerinckii* BA101) that convert biomass-derived sugars to ABE [71]. Commercialization of this technology is in progress, where a bioethanol plant can be readily modified to produce biobutanol. Lignocellulosic sugars can be transformed into petrol fuel-like hydrocarbon fuels via catalysts-assisted dehydrogenation, deoxygenation, hydrogenolysis, and cyclization reactions [72]. In the presence of hydrogen gas, pyrolysis of algal biomass at 300–400 °C results in predominantly (i.e., 85%) hydropyrolysis oil that can be refined to quality biofuels [73]. If water-based slurry containing 20% oil-rich algal biomass is hydrothermally treated at 300–350 °C under 150–200 atm pressure, an organic liquid will be generated for further upgrading to drop-in biofuels [74]. Using particular catalysts such as ruthenium diphosphine, ethanol can be efficiently upgraded to *n*-butanol [75]. Another emerging biofuel technology is Syngas to Gasoline Plus (STG+), through which biomass-derived syngas is transformed into gasoline via a series of catalysts-assisted reactions: $CO+2H_2 \rightarrow CH_3OH$ (methanol); $2CH_3OH \rightarrow CH_3OCH_3$ (dimethyl ether)$+H_2O$; $nCH_3OCH_3 \rightarrow C_6$–$C_{10}$ hydrocarbons (gasoline) [76]. Overall, production and utilization of drop-in biofuels is a future renewable fuel development direction. Intensive research is needed to optimize these technologies for cost-effectiveness and economic viability.

4. Development and utilization of gaseous biofuels

4.1. Biogas

Natural gas, consisting of 95% methane (CH_4) and 5% ethane (C_2H_6), propane (C_3H_8), butane (C_4H_{10}), nitrogen (N_2), and carbon dioxide (CO_2), is a gaseous fossil fuel formed from buried plants and animals that experienced great heat and pressure over thousands of years. The energy content of natural gas is 38.2 MJ m^{-3} (1027 Btu ft^{-3}) or 53.5 MJ kg^{-1} (The density of natural gas is 0.717 kg m^{-3} at standard temperature (0 °C) and pressure (1 atm)). It was in 1821 that natural gas was initially exploited at commercial scale in Fredonia, NY for lighting [77]. Today natural gas is extensively used for cooking, heating, transportation, electricity generation, and industrial manufacturing. In 2011 the global natural gas production reached 4096 billion N m^3 (N: standard conditions with 0 °C temperature and 1 atm pressure), with the U.S. as the largest producer accounting for nearly 20% of the production [78].

Biogas is a renewable gaseous fuel alternative to natural gas. It is generated by anaerobic digestion of organic wastes. Raw biogas consists of 60–65% of methane (CH_4), 30–35% of CO_2, and small percentages of water vapor, H_2, and H_2S [79]. After purification to remove CO_2, H_2S, and other impurities, the upgraded, pipeline-quality biogas (now named *biomethane*) is used as a natural gas substitute.

Organic wastes like plant residues are gradually decomposed by microorganisms to smaller molecules in the natural environment. Under conditions with sufficient air supply, the final decomposition products are generally CO_2 and H_2O. In circumstances like landfill sites and manure lagoons where little oxygen is available, an anaerobic environment forms, and certain microorganisms break down organic residues to CH_4 and CO_2. Microbial decomposition of biomass materials in the absence of oxygen is termed *anaerobic digestion*. Four basic biological steps are involved in anaerobic digestion [79]:

(1) Hydrolysis. Anaerobic bacteria such as *Bacteriocides, Clostridia, Bifidobacteria*, *Streptococci*, and *Enterobacteriaceae* hydrolyze large organic molecules into smaller, simple ones, e.g., carbohydrates to sugars, proteins to amino acids, and lipids to fatty acids.
(2) Acidogenesis. Acidogenic bacteria convert simple organic molecules to carbon dioxide, hydrogen, ammonia, and organic acids.
(3) Acetogenesis. Acetogenic bacteria convert organic acids to acetic acid, along with additional hydrogen, ammonia, and carbon dioxide.
(4) Methanogenesis. Methanogenic bacteria decompose acetic acid to methane and carbon dioxide.

The overall reaction of anaerobic digestion is as:

$$C_6H_{12}O_6 \rightarrow 3CO_2 + 3CH_4 \quad (9)$$

Anaerobic digestion has been utilized by animal farms, wastewater treatment plants, and even individual households to manage waste and/or to produce energy (Fig. 5). Many organic wastes

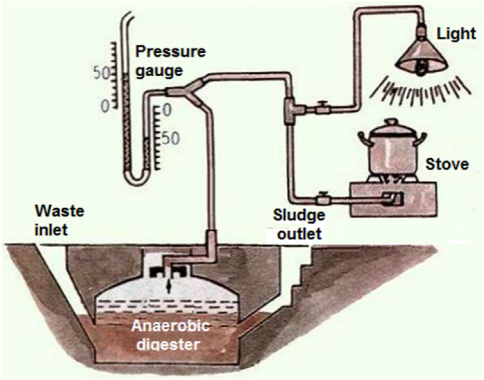


Fig. 5. The structure of a typical anaerobic digester in a rural household in China.

such as livestock manure, food processing waste, kitchen waste, yard trimmings, sewage, and sludge can be used in anaerobic digestion for biogas production. Woody waste should be excluded, as most anaerobic microorganisms cannot degrade lignin [79]. Generally, organic waste in water slurry is placed in an air-tight container called "anaerobic digester" (Fig. 5). After 7–14 days biogas will start to generate. Depending on the temperature and waste types, 30–60 days are generally needed for anaerobic digestion to be complete such that little biogas is further produced [80]. The black, odorous liquid and solids left in the digester need to be cleaned out and replaced by fresh organic wastes. Biogas potential of a particular waste feedstock can be estimated by the Buswell Equation [81]:

$$C_cH_hO_oN_nS_s + 1/4(4c-h-2o+3n+2s)H_2O \rightarrow 1/8(4c+h-2o-3n-2s)CH_4 + 1/8(4c-h+2o+3n+2s)CO_2 + nNH_3 + sH_2S \quad (10)$$

Typically, one ton biowaste (dry weight) would yield 120 m^3 of biomethane that can produce net 200 kW h of electricity [82].

Many factors influence anaerobic digestion and biogas generation. In the digester organic waste should exist in water slurry that contains 15–40% solids. The C/N ratio of the slurry should be less than 24:1 so microorganisms can obtain enough nitrogen to grow. The pH of the slurry should be within 6.5–8.0. Anaerobic microorganisms are active at temperature 25–60 °C (77–140 °F). Temperature should be maintained at a mesophilic range (35–40 °C) or a thermophilic range (50–55 °C) to maximize the microbial activity [82]. In winter time with ambient temperature less than 10 °C or a hot environment at above 70 °C, anaerobic digestion nearly stops [83]. It also takes time for anaerobic microorganisms to establish a fully effective population in an anaerobic digester. A common practice is to introduce anaerobic microorganisms by "seeding" digesters with sewage sludge or cattle slurry that already contain anaerobic microorganism populations [84]. Modern steel-plastic digesters usually have self-heating and self-mixing functions, able to maintain high biogas productivity during the winter time [79].

Biogas was used by human beings to prepare warm bath water as early as in the l0th century B.C. in ancient Assyria. Back to more than 2000 years ago, people in China and India already practiced anaerobic digestion of animal manure for a flammable gas [85]. In 1808, the English chemist Sir Humphry Davy determined that the flammable gas from cattle manure ponds was methane. The first-recorded anaerobic digestion plant was constructed in Bombay, India in 1859. Biogas was collected from a sewage treatment facility to light street lamps in Exeter, England in 1895. The U.S. scientist A.M. Buswell and other colleagues studied anaerobic digestion as a science in the 1930s to select best anaerobic bacteria and digestion conditions for promoting methane production [86]. In 1950s, China built 3.5 million family-sized, low-technology anaerobic digesters in the rural area to provide biogas for cooking and lighting (Fig. 5). In 2012, the total number increased to 45 million, of which roughly 65% are in operation [87]. India has more than 4.5 million small-scale anaerobic digesters to produce biogas from manures. There is a trend in these two countries toward using larger, more sophisticated digestion systems with improved biogas productivity and digester cleansing convenience. Europe had 8960 biowaste digesters (predominantly farm-based; 171 industrial-scale ones) in 2012 to convert agricultural, industrial, and municipal wastes to biogas. Germany was the leading country, operating 6800 biogas plants to meet the nation's natural gas demand and generate 18.2 billion kW h of electricity [88]. The U.S. started to install manure-based digester systems on livestock farms to produce biogas in late 1970s, with financial incentives from the federal government. The sector has been developed rather slowly due to challenges in capital investment, technical maintenance, and economic viability. The number of operating farm-biogas plants increased from 25 in 2000 to 176 in 2011, with 15 new installations a year at the present trend. Biogas from the farm digesters provided sufficient heat to the farms and generated 541 million kW h of electricity in 2011 [89]. In addition, 594 U.S. municipal solid waste disposal facilities and 1238 municipal wastewater treatment plants are collecting biogas from the sites, recovering 155 billion MJ of energy for heat and electricity. In 2012, approximately 9.1 billion kW h of electricity were generated by biogas in the U.S. [88].

It is estimated that worldwide 47–95 billion kW h of electricity were generated from biogas in 2012 [90]. China is presently the largest biogas producer, harvesting 6 billion N m^3 of biomethane per annum. Globally, biogas is still in its infant stage, playing a minor role in the overall bioenergy sector. The global potential for biogas, however, is rather impressive. If majority of the biowaste is anaerobically digested to produce biogas, the yield would displace one quarter of the current natural gas consumption and cover 6% of the global primary energy demand. More than 1000 billion N m^3 of biomethane could be harvested annually assuming the available portion of the world agricultural byproducts (manures, crop residues) and domestic wastes (municipal organic solid refuse, food processing waste, and sewage sludge) were processed by anaerobic digestion [91]. With the significant increases in public awareness of anaerobic digestion and biogas and the availability of supporting structure and technology, biomethane is showing a steady growth into the future, particularly in China and Germany.

4.2. Syngas

Syngas is another gaseous biofuel produced from gasification or pyrolysis of plant materials. Chemically syngas consists of 30–60% CO, 25–30% H_2, 5–15% CO_2, 0–5% CH_4, and lesser portions of water vapor, H_2S, COS, NH_3, and others, depending on the feedstock types and production conditions [92].

Gasification is commercially practiced to produce syngas. In the operation, carbon-rich materials such as coal, petroleum, natural gas, and dry plant biomass is rapidly heated to above 700 °C in the high temperature (e.g., 1200 °C) combustion chamber of a gasifier and partially burned in the presence of controlled air flow to yield syngas (Fig. 6). In the gasifier, wood biomass experiences three thermal transformation phases: dehydration, pyrolysis, and partial oxidation. In the initial dehydration phase, air-dry biomass swiftly loses its moisture before its temperature reaches 200 °C. Pyrolysis begins as the temperature increases, and biomass is converted to char and vapor. In the presence of O_2, char is partially oxidized to generate CO and CO_2, while the vapor is combusted to CO_2 and H_2O. As the hot char particulates, CO, CO_2, H_2O rise in the combustion chamber, further reactions occur that char is oxidized by CO_2 to yield CO, or by H_2O to yield CO and H_2, and CO reacts with H_2O to produce CO_2 and H_2. The mixture of CO, H_2, and CO_2 is then recovered as syngas (Fig. 6). The major reactions can be simplified as follows [92]:

$$\text{Wood} \rightarrow \text{char} + \text{vapor} \quad (11)$$

$$\text{Char} + O_2 \rightarrow CO + CO_2 \quad (12)$$

$$\text{Vapor} + O_2 \rightarrow CO_2 + H_2O \quad (13)$$

$$\text{Char} + CO_2 \rightarrow CO \quad (14)$$

$$\text{Char} + H_2O \rightarrow CO + H_2 \quad (15)$$

$$CO + H_2O \rightarrow CO_2 + H_2 \quad (16)$$

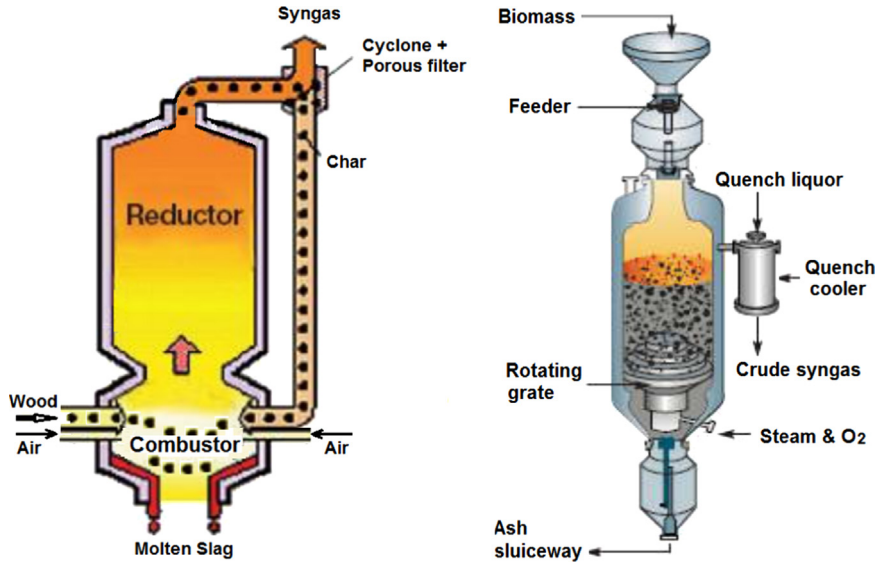


Fig. 6. The structure of a Mitsubishi Heavy Industries gasifier (left) and a Lurgi Dry-Ash gasifier (right) for syngas [92].

The typical gasification of wood to syngas has a carbon conversion rate of 92% (wood to CO, CO_2, CH_4), a hydrogen conversion rate of 71% (wood to H_2, CH_4), and an energy conversion rate of 62% (wood bioenergy to syngas energy). The syngas yield is around 1.2 N m^3 kg^{-1} wood (2.3 N m^3 kg^{-1} if with 48% N_2) [93]. Woody biomass low in nitrogen and ash content is usually used for syngas production. Herbaceous biomass is generally high in ash, especially silica and potassium and tends to cause gasifier slagging.

Crude syngas contains 48% of N_2 and minor amounts of tar, particulates, CO_2, and H_2S. Its energy density (lower heating value) is approximately 5.3 MJ N m^{-3} [93]. The fuel can be directly burned to generate electricity. More commonly the gas is purified and used as a source material for synthesizing transportation fuel, methanol, ethanol, methane, dimethyl ether, and other products. Through the Fischer–Tropsch process, purified syngas has been successfully converted to diesel. Hydrogen separated from syngas was tested to power hydrogen fuel cells for electricity generation and fuel cell electric vehicle propulsion [94]. However, the purification can be costly and energy-consuming. Conventional purification methods include physical and chemical adsorption (e.g., water quenching, amine scrubbing) [95]. Considering that the purified gas has to be reheated prior to use in a gas turbine or chemical synthesis, approaches that can be integrated in gasifiers such as catalytic tar cracking/reforming, CO_2 elimination, and H_2 separation have been researched [92,96].

Due to the stringent requirements for feedstock in moisture content, ash content, size, homogeneity, bulk density, and energy content, worldwide biomass-based gasification is currently at the early demonstration stage [94]. Further research and development is needed to better understand the economics of biomass, improve the cost-effectiveness of feedstock preparation, and achieve the economic viability of bio-syngas production. In 2010, there were 144 commercial gasification plants operating globally with a total capacity of 71,205 MW_{th} (mega watt, thermal). China, Europe, and the U.S. had 56, 42, and 17 plants, respectively. However, only 0.5% of the production was from biomass; the rest was from coal (51%), petroleum (25%), natural gas (22%), and petcoke (1%) [92]. Of the produced syngas, 45% was used to synthesize chemicals, 38% to make transportation fuels, 11% to generate electricity, and 6% as a gaseous fuel [92].

5. Biofuel overview and the potential socio-econo-environmental impacts

Global production and utilization of bioenergy is in various solid, liquid, and gaseous forms of biofuels. Advantages and disadvantages of these biofuels are briefed in Table 2. In general, solid biofuels are most available in source materials, most efficient in feedstock energy recovery, and most effective in conversion technology and production cost, but the products are bulky, inconvenient to handle, low in energy density, and only applicable to solid fuel burners. Liquid biofuels are energy-dense, convenient to transport, and can substitute for gasoline and petrol diesel; however, they are low in net energy efficiency, have stringent requirements for feedstocks, and involve complicated conversion technology and high production cost. Gaseous biofuels can be produced from organic waste materials and residues using well-practiced techniques, yet there are fuel upgrading and byproduct disposal challenges.

The global biofuel utilization reached 0.55×10^{20} J yr^{-1} in 2012, accounting for 10% of the world energy consumption and 80% of the total renewable energy production [97]. Bioethanol and biodiesel in particular, are being widely produced to complement the rapidly depleting petroleum reserves. It is predictable that by 2050, bioethanol and biodiesel would be the dominant fuels to power passenger cars and heavy vehicles. Automobiles powered by electricity, natural gas, and hydrogen are emerging, but the currently existing liquid fuel supply systems will restrict them from becoming the main stream.

Intensive production and utilization of biofuels has generated great social, economic, and environmental impacts. The socio-economic impacts include land rights, food security, energy security, economic viability, local prosperity, labor conditions, and policies. The environmental impacts extend to greenhouse gas emissions, biodiversity, land uses, soil conservation, and water resources. The impacts vary with characteristics of the biofuel supply chains, the production system management, and the conditions of production regions [98]. In general, production and utilization of biofuels enhance a nation's energy security and independence, reduce conventional pollutant and greenhouse gas emissions, promote research and development, create job opportunities, and increase farm income. On the other hand, biomass feedstock production requires land, water, fertilizers,

Table 2
Advantages, disadvantages, and 2013 production status of different biofuels.

Biofuels		Advantages	Disadvantages	Annual production
Solid	Firewood	Renewable, readily available, cheap, most energy efficient	Bulky, low in energy density; high hazardous emissions from incomplete combustion; unsuitable for automated burners	17×10^8 m^3
	Wood chips	More convenient to transport, handle, and store than firewood; lower SO_2 and NO_x emissions than coal upon combustion	Involves chipping cost; tends to decay during storage; bulkier and lower in energy density than coal; ash slagging and boiler fouling; unsuitable for precise combustion	3.5×10^8 t
	Wood pellets	Convenient to transport, handle, and store; low SO_2 and NO_x emissions; suitable for precise combustion	Higher processing cost; lower energy content than coal; only be used in solid fuel burners	20×10^6 t
	Charcoal	Stable, high energy content, clean burning	High production cost; bulk, inconvenient for transport; cannot be used in liquid fuel and gas burners	51×10^6 t
Liquid	Corn/sugar ethanol	Renewable substitute for gasoline; low combustion emissions; existing feedstock production systems	Low net energy efficiency; corrosive to existing gasoline fueling devices; competing with food and feed for source materials	23×10^9 gal
	Cellulosic ethanol	A gasoline alternative from non-food biomass	Low net energy efficiency; not cost-effective	$< 5 \times 10^6$ gal
	Biodiesel	Renewable substitute for petro diesel; existing feedstock production systems	Competes with food production; feedstock is limited to lipids; corrosive to existing diesel fueling devices; substantial processing cost	63×10^8 gal
	Pyrolysis bio-oil	Renewable feedstock; simple conversion technology	Upgrading is needed prior to fuel uses; immature upgrading techniques	Pilot production $< 1 \times 10^6$ gal
	Drop-in fuels	Renewable feedstock; gasoline substitute; compatible with existing fueling systems	Immature, complicated conversion technology; high cost	Early development stage
Gaseous	Biogas	From organic waste and residues, wide feedstock sources; fits the existing natural gas grid	Usually in rural areas; requires intensive feedstock collection and waste disposal	$\sim 25 \times 10^9$ m^3 CH_4
	Syngas	Mature production technology; as feedstock for industrial chemicals	Char and bio-oil as byproducts; stringent requirements for feedstock	4.5×10^{11} m^3

and other resources and may cause land use change, engender additional pressure on water resources (i.e., pollution and overuse), and increase food prices. In certain scenarios, biofuel production may emit more greenhouse gases and consume more fossil energy on an energy-equivalent basis [99]. These negative impacts, however, can be minimized through careful planning and technological advances.

6. Summary and conclusions

Bioenergy in firewood, charcoal, and plant oils was the predominant (i.e., $>95\%$) energy source for human activities such as cooking, heating, lighting, metallurgy, and clay baking before 1840. It was until 1905 that the fossil energy overweighed bioenergy in supporting human life. Over the past one hundred years with the world's population growth and industrial development, the annual global primary energy consumption has increased nearly 10 times, reaching 5.5×10^{20} J in 2010. The current energy demand was met by fossil fuels at 80%, biofuels at 11.3%, uranium at 5.5%, and other renewable source energy at 2.2% [100]. In the past two decades from 1991 to 2010, the global consumption of bioenergy increased steadily from 46.55×10^{18} J to 62.74×10^{18} J. Of all the renewable energy sectors in 2010, bioenergy accounted more than 80% [100].

Biofuels have been developed and utilized in three forms: solid biofuels – firewood, wood chips, wood pellets, and wood charcoal; liquid biofuels – bioethanol, biodiesel, pyrolysis bio-oil, and drop-in transportation fuels; and gaseous biofuels—biogas and syngas. Direct combustion is the most efficient method to utilize bioenergy. As such, solid biofuels, especially firewood, have consistently been the top consumed biofuels. Nearly 40% of the world population relies on firewood for cooking and heating. The global production of firewood has remained relatively constant at around 1900 million m³ per annum, though regional flocculation with 4–12% decreases in North America, Asia-Pacific, and Europe and 3–5% increases in Africa, Latin America, and the Caribbean occurred in the past decades [18]. Wood chips have been increasingly used in co-firing for electricity generation since the beginning of the 21st century. The world electricity generation from wood chips is expected to double from 70 GW in 2010 to 145 GW in 2020. Due to the handling convenience and lower moisture content, wood pellets become more and more popular in residential house heating in developed countries. The global production of wood pellets is projected to expand from 15.4 million tons in 2010 to 45.2 million tons in 2020. Biomass is also the sole source for renewable transportation fuels. The global consumption of bioethanol increased from 4.5 billion gallons in 2000 to 21.8 billion gallons in 2012. However, the bioethanol was predominantly made from starch/sugar-rich food crops such as corn and sugar cane. The tremendous technological advances in converting lignocellulosic biomass into simple sugars, specifically in feedstock pre-treatment and in industrial enzyme preparation, eventually realized the commercial-scale production of second-generation bioethanol from crop residues in 2013. It is expected that the global bioethanol production and utilization will increase significantly in the soon future. In the meanwhile, the world production of biodiesel increased from 213 million gallons in 2000 to 5670 million gallons in 2012. The further increase, however, depends on development of new oil sources, such as cost-effective algal cultivation and oil extraction. Substantial attempts were also made to produce liquid hydrocarbon fuels from plant biomass by pyrolysis. More research is needed to develop practical techniques for upgrading crude pyrolysis bio-oil to quality transportation fuels. Other drop-in fuels such as biobutanol and sugar hydrocarbons are being developed to overcome the infrastructure-incompatibility drawbacks of bioethanol and biodiesel. The technologies are yet in the infant and

early demonstration stages. Renewable bioenergy has also been recovered from organic wastes by anaerobic digestion of animal manures, municipal organic refuse, sewage sludge, and food-processing waste for biogas. With the technology development, biogas may displace one quarter of the current global natural gas consumption in the future. In addition, wood has been tested via gasification to produce syngas that can be used for synthesizing industrial chemicals or for electricity generation. Nevertheless, the technology is not economically advantageous at the current fossil fuel prices.

It is evident that the global consumption of bioenergy will continue to increase under the influence of present renewable energy and climate change policies. Biopower (by co-firing), lignocellulosic bioethanol, and biogas will be the most prosperous bioenergy and renewable sectors with highest growing potential in the soon future. To achieve the pre-set goal of carbon emission reductions in the energy sector, the World Energy Council predicted that the bioenergy consumption would increase three-fold to 2050 relative to the 2010 level [101]. There is strong indication that bioenergy will meet 30% of the whole world's energy demand by 2050.

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